A Counterexample to a Character Count under Coprime Action

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Abstract

Let a finite group A act coprimely on a finite group G, and set $C = C_G(A)$. For

$$N_A(G) = \{\chi \in \mathrm{Irr}(G) : \chi^a = \chi \text{ for all } a \in A,\ \chi(c) \neq 0 \text{ for all } c \in C\},$$

we consider whether $|N_A(G)| = |C/C'|$ always holds. We construct a counterexample with $A \cong C_{21}$ and $G \cong C_2^{128} \rtimes C_2^7$. Here $C \cong C_2^{12}$, while the little-groups method identifies the A-invariant irreducible characters with the 4096 A-fixed Boolean functions on $\mathbf{F}_4^2 \oplus \mathbf{F}_8$. Their values on C reduce to correlations of six-bit radial functions on $\mathbf{F}_4^2$, and the resulting binary systems give

$$|N_A(G)| = 1728 < 4096 = |C/C'|.$$

Three exact computer calculations independently check the count but are not used in the proof.

1 Introduction

Let A act by automorphisms on a finite group G, with $(|A|, |G|) = 1$, and set $C = C_G(A)$. We consider whether the number of A-invariant irreducible characters of G that are nonzero on C must equal the number of linear characters of C:

$$|N_A(G)| \stackrel{?}{=} |C/C'|, \qquad N_A(G) = \{\chi \in \mathrm{Irr}^A(G) : \chi(c) \neq 0 \text{ for all } c \in C\}. \tag{1}$$

This question, proposed by G. Navarro, appears as Problem 21.100 in the 21st issue of the Kourovka Notebook [1].

We construct a cyclic group A of order 21 acting on a 2-group $G = B \rtimes V$, where $V \cong C_2^7$ and B is the additive group of all Boolean functions on V. The action on V has no nonzero fixed point and has twelve orbits, giving $C_G(A) = B^A \cong C_2^{12}$.

The little-groups method gives an explicit parameterization of $\mathrm{Irr}(G)$. Each A-stable translation orbit in B has a unique A-fixed representative, and the only invariant character of its translation stabilizer is the trivial character. Thus $\mathrm{Irr}^A(G)$ is indexed by B^A. Writing an A-fixed Boolean function on $\mathbf{F}_4^2 \oplus \mathbf{F}_8$ as a pair of six-bit radial functions on $\mathbf{F}_4^2$ reduces the character values on C to two binary correlation functions. The resulting zero criterion gives $|N_A(G)| = 1728$.

The construction is not claimed to have minimal order, and no claim is made about operator groups subject to additional restrictions.

2 The construction and its fixed subgroup

Let

$$U = \mathbf{F}_4^2, \qquad W = \mathbf{F}_8, \qquad V = U \oplus W$$

as additive groups. Choose $\omega \in \mathbf{F}_4^\times$ of order 3 and $\zeta \in \mathbf{F}_8^\times$ of order 7. Define

$$T(x_1, x_2, y) = (\omega x_1, \omega x_2, \zeta y).$$

Then T is $\mathbf{F}_2$-linear and has order 21. Moreover,

$$C_V(T) = 0, \tag{2}$$

because $\omega - 1$ and $\zeta - 1$ are nonzero field elements.

Let

$$B = \{f : V \to \mathbf{F}_2\}$$

with pointwise addition. For $t \in V$, define

$$(\tau_t f)(x) = f(x+t).$$

The additive group V acts on B through these translations. Put

$$G = B \rtimes V, \qquad (f,t)(g,s) = (f + \tau_t g, t+s).$$

Thus $|G| = 2^{128} 2^7 = 2^{135}$.

Let $A = \langle a \rangle \cong C_{21}$. Define

$$a \cdot (f,t) = (f \circ T^{-1}, Tt).$$

This is an automorphism of G, since

$$(\tau_t f) \circ T^{-1} = \tau_{Tt}(f \circ T^{-1}).$$

Its restriction to V is T, so the action is faithful. Since $|A| = 21$ and $|G|$ is a power of 2, the action is coprime.

Proposition 2.1. *For the action above,*

$$C_G(A) = B^A \cong C_2^{12}.$$

In particular,

$$|C_G(A)/C_G(A)'| = 4096.$$

Proof. If $(f,t) \in G$ is fixed by a, then $Tt = t$. Equation (2) gives $t = 0$, so $C_G(A) = B^A$.

It remains to count the A-orbits on V. The zero vector is one orbit. The nonzero vectors in U split into the five sets $L_i^\times$, where $L_1, \ldots, L_5$ are the one-dimensional $\mathbf{F}_4$-subspaces of U; each has size 3 and is one orbit under multiplication by ω. The set $\{0\} \times W^\times$ is one orbit of size 7. Finally, each

$$L_i^\times \times W^\times$$

is an orbit of size 21, since the powers of (ω, ζ) run through $C_3 \times C_7$.

The orbit sizes are therefore

$$1, 3, 3, 3, 3, 3, 7, 21, 21, 21, 21, 21.$$

An element of B is fixed by A exactly when it is constant on each of these twelve orbits. Hence $B^A \cong C_2^{12}$. This group is abelian, so its derived subgroup is trivial and the final assertion follows. $\square$

3 The invariant irreducible characters

For $u, f \in B$, let

$$\langle u, f\rangle = \sum_{x \in V} u(x)f(x) \in \mathbf{F}_2$$

and define

$$\lambda_u(f) = (-1)^{\langle u,f\rangle}.$$

The map $u \mapsto \lambda_u$ identifies B with $\mathrm{Irr}(B)$.

For $u \in B$, write

$$V_u = \{t \in V : \tau_t u = u\}, \qquad I_u = B \rtimes V_u.$$

The subgroup I_u is the inertia subgroup of λ_u in G. For $\mu \in \mathrm{Irr}(V_u)$, define a linear character of I_u by

$$\theta_{u,\mu}(f, t) = \lambda_u(f)\mu(t).$$

This is well defined because V_u fixes λ_u. Put

$$\chi_{u,\mu} = \mathrm{Ind}_{I_u}^{G} \theta_{u,\mu}.$$

Proposition 3.1. *Choose one u from each translation orbit of V on B. Then*

$$\mathrm{Irr}(G) = \{\chi_{u,\mu} : u \textit{ is a chosen representative}, \ \mu \in \mathrm{Irr}(V_u)\}.$$

Every character in this set has degree $[V : V_u]$.

Proof. The subgroup I_u is normal in G, since it is the inverse image of V_u under $G \to G/B = V$ and V is abelian. Mackey's inner-product formula gives

$$\langle \chi_{u,\mu}, \chi_{u,\nu}\rangle_G = \sum_{gI_u \in G/I_u} \langle \theta_{u,\mu}, \theta_{u,\nu}^{g}\rangle_{I_u}.$$

If $g \notin I_u$, the restrictions of $\theta_{u,\mu}$ and $\theta_{u,\nu}^{g}$ to B are distinct, because g sends λ_u to a different translate. Those terms vanish. The identity coset contributes 1 when $\mu = \nu$ and 0 otherwise. Thus the characters $\chi_{u,\mu}$ are irreducible and are distinct for fixed u. Characters arising from different translation orbits have disjoint sets of constituents on restriction to B, so they are also distinct.

For one translation orbit represented by u, there are $|V_u|$ characters, each of degree $[V : V_u]$. Their contribution to the sum of squared degrees is

$$|V_u|[V : V_u]^2 = |V|\,|V \cdot u|.$$

Summing over all translation orbits in B gives

$$\sum_{u,\mu} \chi_{u,\mu}(1)^2 = |V| \sum_{\mathcal{O} \subseteq B} |\mathcal{O}| = |V||B| = |G|.$$

Since these distinct irreducible characters account for the full sum of squared degrees, they exhaust $\mathrm{Irr}(G)$. □

Lemma 3.2. *Every A-stable translation orbit in B contains exactly one element of B^A.*

Proof. Suppose the translation orbit of u is A-stable. Then

$$a \cdot u = \tau_z u$$

for some $z \in V$, determined modulo V_u. The subgroup V_u is T-invariant. Indeed, if $t \in V_u$, then

$$a \cdot (\tau_t u) = \tau_{Tt}(a \cdot u) = \tau_{Tt+z} u.$$

Since $a \cdot u = \tau_z u$, it follows that $\tau_{Tt} u = u$, and hence $Tt \in V_u$.

The operator T has no nonzero fixed point on V/V_u. If $t + V_u$ is fixed, then $T^i t \equiv t \pmod{V_u}$ for every i. Put

$$s = \sum_{i=0}^{20} T^i t.$$

The vector s is fixed by T, so $s = 0$ by (2). On the other hand, $s \equiv 21t \equiv t \pmod{V_u}$, because V has characteristic 2. Thus $t \in V_u$.

Therefore $T - I$ has trivial kernel on the finite-dimensional space V/V_u, and is invertible. Choose w with

$$(T - I)w \equiv z \pmod{V_u}.$$

Since V has characteristic 2, the congruence gives $(T - I)w + z \in V_u$, and hence $Tw + z \equiv w \pmod{V_u}$. Thus

$$a \cdot (\tau_w u) = \tau_{Tw+z} u = \tau_w u,$$

so the orbit contains an A-fixed element.

If both u and $\tau_t u$ are fixed, then $t + V_u$ is fixed in V/V_u. The preceding argument gives $t \in V_u$, so $\tau_t u = u$. The fixed representative is unique. □

Proposition 3.3. *The A-invariant irreducible characters of G are precisely*

$$\chi_u = \operatorname{Ind}_{B \rtimes V_u}^{G} \widetilde{\lambda}_u, \qquad u \in B^A,$$

where

$$\widetilde{\lambda}_u(f, t) = \lambda_u(f).$$

In particular,

$$|\operatorname{Irr}^A(G)| = 4096.$$

Proof. By Proposition 3.1, an invariant irreducible character must arise from an A-stable translation orbit. Lemma 3.2 supplies a unique representative $u \in B^A$.

For fixed $u \in B^A$, the extension $\widetilde{\lambda}_u$ is A-invariant, and

$$\chi_{u,\mu}^a = \chi_{u,\mu^a}.$$

The parameterization in Proposition 3.1 is injective, so $\chi_{u,\mu}$ is invariant if and only if μ is invariant.

Write $\mu(t) = (-1)^{\ell(t)}$ for an $\mathbf{F}_2$-linear functional ℓ on V_u. Invariance gives $\ell(Tt) = \ell(t)$, so ℓ vanishes on $(T - I)V_u$. By (2), the restriction of $T - I$ to V_u is injective. Since V_u is finite-dimensional, this restriction is invertible. Hence $\ell = 0$ and μ is trivial. The parameterization follows, and Proposition 2.1 gives $|B^A| = 4096$. □

For $u, c \in B^A$, let $x_t = (0,t)$. Since I_u is normal in G and $c \in B \subseteq I_u$, the induced-character formula, with x_t ranging over representatives for G/I_u, gives

$$\chi_u(c) = \sum_{t \in V/V_u} \widetilde{\lambda}_u\big(x_t^{-1}(c,0)x_t\big).$$

Now

$$x_t^{-1}(c,0)x_t = (\tau_{-t}c, 0)$$

and

$$\langle u, \tau_{-t}c\rangle = \langle \tau_t u, c\rangle.$$

Therefore

$$\chi_u(c) = \sum_{t \in V/V_u} (-1)^{\langle \tau_t u, c\rangle}. \tag{3}$$

The summand is constant on cosets of V_u, so

$$|V_u|\chi_u(c) = \sum_{t \in V} (-1)^{\langle \tau_t u, c\rangle}. \tag{4}$$

4 Radial correlations

The action of $\langle \omega \rangle$ on U has six orbits,

$$\{0\}, L_1^\times, \ldots, L_5^\times.$$

A Boolean function on U that is constant on these orbits will be called radial. We encode it by

$$X = (x_0, x_1, \ldots, x_5) = (x_0, x) \in \mathbf{F}_2^6$$

and write

$$\tau(X) = x_0 + \sum_{i=1}^{5} x_i.$$

Lemma 4.1 (Two-slice decomposition). *Every $u \in B^A$ has a unique representation*

$$u(z,w) = \begin{cases} X(z), & w = 0, \\ Y(z), & w \neq 0, \end{cases}$$

where X and Y are radial functions on U.

Proof. The slice at $w = 0$ is radial by invariance under a. If $w \neq 0$, then a^7 fixes w and acts on U as multiplication by $\omega^7 = \omega$, so the slice at w is also radial.

For $w, w' \neq 0$, choose k such that $\zeta^k w = w'$. Invariance gives

$$u(z,w) = u(\omega^k z, w').$$

The slice at w' is radial, so the right side equals $u(z, w')$. Hence all nonzero slices coincide. Uniqueness is immediate. □

We therefore encode $u \in B^A$ by a pair $(X, Y) \in \mathbf{F}_2^6 \times \mathbf{F}_2^6$, and use (E, F) for $c \in B^A$.

For radial functions X, E on U, define

$$K_X(E)(s) = \sum_{z \in U} X(z+s)E(z) \in \mathbf{F}_2.$$

This is again radial.

Lemma 4.2 (Correlation formula)**.** *Let $X = (x_0, x)$ and $E = (e_0, e)$, where $x, e \in \mathbf{F}_2^5$. Put $s_x = \sum_i x_i$, $s_e = \sum_i e_i$, and $\mathbf{1} = (1,1,1,1,1)$. Then*

$$K_X(E) = \left(x_0e_0 + x \cdot e,\ \tau(E)x + \tau(X)e + (s_xs_e + x \cdot e)\mathbf{1}\right). \tag{5}$$

Proof. At $s = 0$,

$$K_X(E)(0) = x_0e_0 + \sum_{i=1}^{5} 3x_ie_i = x_0e_0 + x \cdot e.$$

Fix $s \in L_i^\times$. The terms $z = 0$ and $z = s$ contribute $x_ie_0 + x_0e_i$. The other two nonzero elements of L_i contribute x_ie_i twice and cancel.

For $j \neq i$, the three points of $s + L_j^\times$ lie one each on the three projective lines different from L_i and L_j. Indeed, none lies on L_i or L_j, and two cannot lie on the same third line, since their difference would belong both to that line and to L_j. The contribution of $L_j^\times$ is therefore

$$e_j \sum_{k \neq i,j} x_k.$$

Thus

$$K_X(E)_i = x_ie_0 + x_0e_i + \sum_{j \neq i} e_j \sum_{k \neq i,j} x_k.$$

Expanding this expression in $\mathbf{F}_2$ yields the ith coordinate in (5). ∎

Let $u = (X, Y)$ and $c = (E, F)$. For $t = (s, v) \in U \oplus W$, put

$$h_{u,c}(t) = \langle \tau_t u, c \rangle.$$

If $v = 0$, then

$$H_0(s) := h_{u,c}(s, 0) = K_X(E)(s) + K_Y(F)(s). \tag{6}$$

If $v \neq 0$, then

$$H_1(s) := h_{u,c}(s, v) = K_Y(E)(s) + K_X(F)(s). \tag{7}$$

For (7), the W-coordinates 0 and v interchange the two slices. The other six nonzero coordinates contribute the same term six times and cancel in $\mathbf{F}_2$. In particular, H_1 is independent of the chosen nonzero v.

Set

$$P = X + Y, \qquad Q = E + F.$$

Bilinearity gives

$$H_0 + H_1 = K_P(Q), \tag{8}$$

$$H_0 = K_P(E) + K_Y(Q). \tag{9}$$

For a radial function $H = (h_0, h_1, \ldots, h_5)$, define

$$S(H) = (-1)^{h_0} + 3\sum_{i=1}^{5}(-1)^{h_i}.$$

Equation (4) becomes

$$|V_u|\chi_u(c) = S(H_0) + 7S(H_1). \tag{10}$$

5 The exact zero criterion

Let

$$\delta = (1,0,0,0,0,0), \qquad J = (1,1,1,1,1,1).$$

For $R \subseteq \{1,\ldots,5\}$, let

$$\rho_R = (0, \mathbf{1}_R),$$

where $\mathbf{1}_R$ is the indicator vector of R.

Lemma 5.1 (Zero patterns)**.** *Equation* (10) *vanishes if and only if*

$$(H_0, H_1) = (\delta + \varepsilon J, \rho_R + \varepsilon J) \tag{11}$$

for some $\varepsilon \in \mathbf{F}_2$ and some three-element subset $R \subseteq \{1,\ldots,5\}$.

Proof. Write the signs associated with H_0 as $a, b_1, \ldots, b_5 \in \{\pm 1\}$ and those associated with H_1 as $d, e_1, \ldots, e_5 \in \{\pm 1\}$. The vanishing equation is

$$a + 3\sum_{i=1}^{5} b_i + 7d + 21\sum_{i=1}^{5} e_i = 0.$$

The last sum is odd. The absolute value of the preceding terms is at most $1 + 15 + 7 = 23$, so $\sum e_i = \pm 1$.

If $\sum e_i = -1$, the remaining terms must total 21. This forces $a = -1$, $b_i = 1$ for all i, and $d = 1$. Exactly three of the e_i are negative. This is the pattern (δ, ρ_R) with $|R| = 3$. The case $\sum e_i = 1$ is its simultaneous sign reversal, obtained by adding J to both binary vectors. $\square$

A zero therefore requires

$$K_P(Q) = H_0 + H_1 = \delta + \rho_R \tag{12}$$

for a three-set R.

Write

$$P = (p_0, p), \qquad p \in \mathbf{F}_2^5.$$

Lemma 5.2. *If $\tau(P) = 1$, then K_P is invertible on the six-dimensional space of radial functions on U.*

Proof. Identify all functions on the additive group $U \cong C_2^4$ with the group algebra $\mathbf{F}_2[U]$. Since every element of U is its own inverse, the coefficient of s in the product of P and Q is $K_P(Q)(s)$. Thus K_P is multiplication by P.

The augmentation ideal of $\mathbf{F}_2[U]$ is nilpotent. For example, after choosing a basis $g_1, \ldots, g_4$ of U and setting $z_i = g_i + 1$, one has

$$\mathbf{F}_2[U] \cong \mathbf{F}_2[z_1, z_2, z_3, z_4]/(z_1^2, z_2^2, z_3^2, z_4^2).$$

An element of augmentation 1 is therefore $1+n$ with n nilpotent, and is a unit. The augmentation of the radial function P is

$$p_0 + 3\sum_{i=1}^{5} p_i = \tau(P).$$

Hence P is a unit when $\tau(P) = 1$.

Multiplication by ω induces an automorphism of the group algebra that fixes P. It also fixes the unique inverse of P. The inverse is therefore radial, so multiplication by P restricts to an invertible map on the radial subspace. □

Proposition 5.3 (Complete zero criterion)**.** *Let $u = (X, Y) \in B^A$, put $P = X + Y = (p_0, p)$, and write $Y = (y_0, y)$. If $\tau(P) = 0$, set*

$$r = p + p_0\mathbf{1}.$$

Then χ_u vanishes at some element of C if and only if at least one of the following holds:

(i) $\tau(P) = 1$*;*

(ii) $\tau(P) = 0$*,* $\mathrm{wt}(r) = 2$*,* $\tau(Y) = 1$*, and*

$$r \cdot y = 1 + p_0.$$

Proof. Suppose first that $\tau(P) = 1$. Lemma 5.2 allows us to choose, for any three-set R,

$$Q = K_P^{-1}(\delta + \rho_R)$$

and then

$$E = K_P^{-1}(\delta + K_Y(Q)).$$

Equations (8) and (9) give $H_0 = \delta$ and $H_1 = \rho_R$. Lemma 5.1 yields a zero.

Assume now that $\tau(P) = 0$. For $Q = (q_0, q)$, formula (5) simplifies to

$$K_P(Q) = \alpha J + \beta(0, r), \tag{13}$$

where

$$\alpha = p_0 q_0 + p \cdot q, \qquad \beta = \tau(Q).$$

Indeed, the first coordinate is α, while the five line coordinates are $\alpha\mathbf{1} + \beta r$. The parameters satisfy

$$\alpha + p_0\beta = r \cdot q. \tag{14}$$

If $r \neq 0$, any prescribed pair (α, β) occurs: choose q with $r \cdot q = \alpha + p_0\beta$ and then put $q_0 = \beta + \sum_i q_i$.

The target in (12) has first coordinate 1 and a line part of weight 3. A vector in (13) with first coordinate 1 has line part $\mathbf{1}$ or $\mathbf{1} + r$, of weights 5 and $5 - \mathrm{wt}(r)$. Since r has even weight, the target occurs precisely when $\mathrm{wt}(r) = 2$. In that case R is the complement of $\mathrm{supp}(r)$, and (12) is equivalent to

$$\tau(Q) = 1, \qquad r \cdot q = 1 + p_0. \tag{15}$$

We next determine when E can be chosen in (9). Because J belongs to the image of K_P, the parameter ε in (11) is immaterial. The required solvability is equivalent to

$$\delta + K_Y(Q) \in \langle J, (0, r)\rangle. \tag{16}$$

Write $t = \tau(Y)$ and $\sigma = \sum_i q_i$. Since $\tau(Q) = 1$, one has $q_0 = 1 + \sigma$. Substituting (5) into (16), and subtracting the multiple of J determined by the first coordinate, gives

$$y + tq + (1 + y_0 + t\sigma)\mathbf{1} \in \{0, r\}. \tag{17}$$

If $t = 0$, the coordinate sum of the left side of (17) is

$$\sum_i y_i + 1 + y_0 = 1,$$

since $\sum_i y_i = y_0$. Both 0 and r have even coordinate sum, so $t = 0$ is impossible. Thus $\tau(Y) = 1$.

Taking the dot product of (17) with r gives $r \cdot y = r \cdot q$, because $r \cdot \mathbf{1} = r \cdot r = 0$ in $\mathbf{F}_2$. Equation (15) now gives

$$r \cdot y = 1 + p_0.$$

Conversely, suppose $\tau(Y) = 1$ and $r \cdot y = 1 + p_0$. Define

$$q = y + (1 + y_0)\mathbf{1}, \qquad q_0 = 1.$$

Since $\sum_i y_i = 1 + y_0$, one has $\sum_i q_i = 0$ and hence $\tau(Q) = 1$. Also $r \cdot q = r \cdot y = 1 + p_0$, because $r \cdot \mathbf{1} = 0$. Equation (15) holds, and the left side of (17) is zero. Thus an appropriate E exists, and the character has a zero. □

6 Counting the zero-free characters

Theorem 6.1. *For the action constructed in Section 2,*

$$|N_A(G)| = 1728 \qquad \textit{and} \qquad |C_G(A)/C_G(A)'| = 4096.$$

In particular, the equality in (1) *does not hold for all coprime actions.*

Proof. The second equality is Proposition 2.1. To obtain the first, count the pairs (X, Y) that fail both conditions in Proposition 5.3.

The change of variables

$$(X, Y) \longleftrightarrow (P, Y), \qquad P = X + Y,$$

is a bijection on $\mathbf{F}_2^6 \times \mathbf{F}_2^6$. There are 32 choices of P with $\tau(P) = 1$, and Proposition 5.3 shows that all 64 choices of Y then give vanishing characters.

Suppose $\tau(P) = 0$. The map

$$P = (p_0, p) \longmapsto (p_0, r), \qquad r = p + p_0\mathbf{1},$$

is a bijection between such P and pairs in which $p_0 \in \mathbf{F}_2$ and $r \in \mathbf{F}_2^5$ has even weight. There are

$$2, \qquad 2\binom{5}{2} = 20, \qquad 2\binom{5}{4} = 10$$

choices with $\mathrm{wt}(r) = 0, 2, 4$, respectively.

If $\mathrm{wt}(r) = 0$ or 4, every one of the 64 choices of Y is zero-free. Fix a critical P with $\mathrm{wt}(r) = 2$. Vanishing is equivalent to the two affine equations

$$\tau(Y) = 1, \qquad r \cdot y = 1 + p_0.$$

These equations are independent: the first involves y_0, while the second does not and $r \neq 0$. Exactly $2^{6-2} = 16$ of the 64 choices of Y satisfy both equations, leaving 48 zero-free choices. Therefore

$$\begin{aligned}|N_A(G)| &= 2 \cdot 64 + 10 \cdot 64 + 20 \cdot 48 \\ &= 128 + 640 + 960 \\ &= 1728.\end{aligned}$$

Since $1728 < 4096$, the equality in (1) fails. □

Remark 6.2 (A single explicit zero). The strict inequality can be seen without the full count. Let $u = \delta_0$ be the indicator of $0 \in V$. Then $V_u = 0$, and $\chi_u = \mathrm{Ind}_B^G \lambda_u$ is an A-invariant irreducible character of degree 128. Choose three distinct projective lines L_1, L_2, L_3 in U and let c be the indicator of

$$\{0\} \cup \bigcup_{i=1}^{3} (L_i^\times \times W^\times).$$

This set is A-stable and has size $1 + 3 \cdot 21 = 64$, so $c \in C$ and

$$\chi_u(c) = \sum_{t \in V} (-1)^{c(t)} = 64 - 64 = 0.$$

The full analysis above gives the exact count 1728.

7 Verification and limitations

Three exact Python implementations were used as independent checks: a full integer Walsh-transform computation, a direct computation in $\mathbf{F}_4^2 \oplus \mathbf{F}_8$, and a binary linear-algebra test based on the twenty zero patterns in Lemma 5.1. Each returns 1728 zero-free characters and 2368 characters with a zero. These computations are not used in the proof. The source code is available at `https://github.com/erichou1/coprime-action-character-count-counterexample`.

The example is not claimed to be minimal. It uses a cyclic operator group of order 21, and the argument does not address versions of the question with additional restrictions on A or G.

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